THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Measurement of Length and Motion, Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1000 metres	conversion
2	**(d)** measuring tape	flexible
3	**(b)** millimetre	small lengths
4	**(c)** a reference point	frame of reference
5	**(b)** oscillatory (periodic)	to and fro
6	**(c)** opaque	e.g. wood
7	**(d)** Frosted glass	light passes partly
8	**(b)** Sugar	dissolves
9	**(c)** Wooden block	less dense
10	**(d)** classification	sorting

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Linear, circular, rotational, oscillatory, periodic, random.	1+1
12	A car on a straight road; a spinning top.	1+1
13	Allows heat or electricity to pass; copper and iron.	1+1
14	Does not allow heat or electricity to pass; wood and plastic.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rapid to-and-fro motion; guitar string (3).	1+2
16	Has tiny holes (3).	1+2
17	10 mm = 1 cm; 100 cm = 1 m; 1000 m = 1 km (3).	1+2
	OR: Wheels rotate, pedals move in circles, the cycle moves linearly (3).	1+2
18	Windows; frosted bathroom glass; walls (3).	1+2
	OR: Solids have a fixed shape; liquids take the shape of container (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 0.8 km/min (1). (ii) 48 km/h (1). (iii) Linear (1). (iv) Using a speedometer and a clock (2).	1+1+1+2
	OR: (i) About own axis vs around another body (3). (ii) Earth spinning and orbiting the Sun (2).	3+2
20	(i) Salt and sugar dissolve; sand settles (1). (ii) Salt, sugar (1). (iii) Evaporation (1). (iv) Solubility is a property (2).	1+1+1+2
	OR: (i) Origin (2). (ii) Examples (2). (iii) Natural materials (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** periodic motion	e.g. clock hands
2	**(d)** curvilinear motion	along a curved path
3	**(a)** Kilogram	mass unit
4	**(d)** Stopwatch	time measurement
5	**(b)** 100	SI scale
6	**(c)** gas	properties
7	**(d)** ductility	e.g. copper wire
8	**(d)** Copper	element
9	**(c)** Plastic is an insulator	safety
10	**(d)** hard and strong	property

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Motion that repeats after equal time intervals; a clock hand.	1+1
12	It changes direction unpredictably.	1+1
13	Metals are good conductors of heat.	1+1
14	It is a poor conductor so the handle stays cool.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The zero is lost and reading may be wrong (3).	1+2
16	Appearance, hardness, solubility, floating (3).	1+2
17	Measure a stack of many sheets and divide by their number (3).	1+2
	OR: 72 mm (1.5); 0.56 m (1.5).	1½+1½
18	Glass lets us see outside (3).	1+2
	OR: Metals conduct heat (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Using steps, tape, rope (2). (ii) Compare (2). (iii) Standard units are better (1).	2+2+1
	OR: (i) 2 s (1). (ii) Oscillatory/periodic (1). (iii) The time period increases (1). (iv) Reduces error (2).	1+1+1+2
20	(i) Has mass and space (1). (ii) Inflate and weigh (2). (iii) Air occupies space (2).	1+2+2
	OR: (i) Solid, liquid, gas (1). (ii) Heating (1). (iii) No (1). (iv) Solid (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Measurement of Length and Motion, Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 10 millimetres	conversion
2	**(d)** directly above the mark	avoid parallax error
3	**(c)** So that everyone gets the same measurement	uniformity
4	**(d)** position changes with time	definition
5	**(b)** oscillatory motion	back and forth
6	**(a)** translucent	e.g. tracing paper
7	**(d)** lustre	property
8	**(b)** Sand	does not dissolve
9	**(c)** Iron nail	denser than water
10	**(c)** Solid	definite shape

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Handspan, foot, cubit.	1+1
12	148 mm.	1+1
13	Malleability: beaten into sheets; ductility: drawn into wires.	1+1
14	Transparency.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A fixed place to describe position; school gate, tree (3).	1+2
16	Mass is the amount of matter; volume is the space occupied (3).	1+2
17	Tape; ruler; odometer or tape (3).	1+2
	OR: It repeats after equal times; oscillatory (3).	1+2
18	Conducting, heat conduction, flexibility (3).	1+2
	OR: It is malleable and non-toxic (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Accuracy (2). (ii) Height, cloth, distance (2). (iii) cm, m, km (1).	2+2+1
	OR: (i) Distances from two lines (3). (ii) Correct placement (2).	3+2
20	(i) Scratch test (3). (ii) Chalk < coal < iron (2).	3+2
	OR: (i) Plastic, metal, glass (2). (ii) Waterproof, conducting, transparent (3).	2+3

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** circular (periodic) motion	go round
2	**(a)** vibration	oscillatory
3	**(c)** 125 mm	×10
4	**(d)** time/period of oscillation	history
5	**(c)** the door as the reference point	position
6	**(c)** mixture of gases	matter around us
7	**(b)** Light	has no mass
8	**(c)** good conduction of heat	use of material
9	**(c)** has small holes (is porous)	property
10	**(d)** texture	touch

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Metre; centimetre/millimetre.	1+1
12	To measure curved surfaces, such as waist and long lengths.	1+1
13	Feel of a surface; sandpaper is rough, glass is smooth.	1+1
14	To study them easily and select suitable ones.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	She is 50 m north and 20 m east of the gate (3).	1+2
16	Volume stays the same; shape changes (3).	1+2
17	Sundial uses the Sun's shadow; a clock uses mechanical or electrical movement (3).	1+2
	OR: Any five correctly classified (3).	1+2
18	Floating: cork, leaf; sinking: coin, nail (3).	1+2
	OR: Safe, durable and heat-conducting (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Sundials, handspans (3). (ii) For accuracy and trade (2).	3+2
	OR: (i) Distance is the total path; position change is the straight line between start and end (3). (ii) Walk 3 m east and 4 m north: path 7 m, straight line 5 m (2).	3+2
20	(i) To choose correct materials (3). (ii) Engineers, tailors (2).	3+2
	OR: (i) Having tiny holes (1). (ii) Sponge, brick (2). (iii) Absorb liquids (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.